Why did this paper cite that one?

Typed citation edges for a research-literature knowledge graph

Callum Young · *affiliation to be filled in*

809 citation sites · 3-class intent taxonomy · **95% on a blind gold set** (macro-F1 0.892)

1 What Prior is, and why citations

Prior reads a corpus of research papers and builds a map of them. It pulls out each paper's *contributions* — the specific things that paper claims to have done — and draws typed, directed relations between the contributions of different papers, so you can ask *"what is actually known about X?"* and get an answer grounded in the literature rather than a plausible-sounding summary.

The map has a weak link. Those relations are drawn from **semantic similarity plus an LLM's reading of two contribution statements**. The model never sees whether one paper actually cited the other, or what it said when it did. The result is a map that agrees with itself: **70% of edges are `supports`**, and `contradicts` — the label you would most want a literature map for — has been measured at about **36% precision**.

Meanwhile the citation record sits unused. And a citation is a different kind of object: p_1 citing p_2 is a **fact**; an LLM saying their ideas "build on" each other is an **assertion**.

But raw citations are blunt. In the graph, p_1 *extending* p_2's method, p_1 *mentioning* p_3 in passing, and p_1 *name-dropping* p_4 out of obligation are all the same edge with the same weight. **The reason for a citation is the missing variable.**

So: type the citation edges (p_1, p_2, t), then use those facts to discipline the assertions.

[screenshot pending]
`figures/atlas.png`
The Prior atlas — zoomed out, topic clusters visible

Fig 1. Prior's semantic map of the 152-paper corpus. Relations here are drawn without ever looking at a citation.

2 The taxonomy, and why ours

We started from **RefWarden**, a citation-verification tool, which asks whether a cited paper `exists`, whether it `supports` the claim it is attached to, and whether it is `obligatory` or merely `helpful`. Good questions — but they all ask *is this citation sound?* Prior needs to know *why the citation is there at all.*

So we added a primary **intent** axis, keeping support and priority as secondary stamps:

- `background` — cited as context, prior art, or a passing mention.
- `uses_extends` — the citing paper actually adopts or builds on the target.
- `compares_contrasts` — the citing paper sets its own work *specifically* against the target.

The selection rule: a class only earns its place if the downstream system would *act differently* on it.

That single test does all the work. ACL-ARC's `uses` and `extends` collapse into one class, because Prior treats them identically. Semantic Scholar's `result` is dropped — ambiguous, and it triggers no distinct action. `exists` is excluded because every cited paper is already in the corpus, so it is trivially yes (parked for the hallucinated-citation demo on AI-generated papers).

And why we could not just use an off-the-shelf label set: Semantic Scholar has no contrast class at all. Contrast is the single most valuable signal for Prior, because `contradicts` is exactly the relation the semantic graph gets wrong. The standard taxonomy cannot express the one thing we most need.

Intent is also not redundant with verification: **65%** of `compares_contrasts` sites still `support` their local claim — so the support axis genuinely cannot see intent.

scite	ACL-ARC	Sem. Scholar	Ours
mentioning	background	background	background
supporting	uses, extends	method	uses_extends
contrasting	compare/ contrast	— *none* —	compares_contrasts
—	motivation, future work	result	*dropped*

the gap that forced our own judge ↗

Fig 2. Four taxonomies. Classes merge when the downstream action is the same, and drop when there is no distinct action. Semantic Scholar's missing contrast class is why an off-the-shelf label set was not an option.

3 The pipeline

The corpus is **152 papers** on AI-for-science, already carrying **581 contributions** and **989 semantic edges** from Prior's existing pipeline.

For each paper we extract the bibliography from its arXiv LaTeX source, resolve every reference back to a paper in the corpus, and pull the **passage around the citation**, marking the reference being typed as `[CITED:TARGET]` and its neighbours as `[CITED]`. Intra-corpus citations only.

That gives **525 citation edges across 809 claim-sites**, later extended to **1,103 sites over 643 paper-pairs**. A **site** is one (edge, passage) pair; an **edge** is one citing→cited paper pair, rolled up from its sites.

Everything ran on a Claude Code subscription rather than a metered API — which is why every runner is checkpointed, resumable, and stops cleanly when the usage window closes.

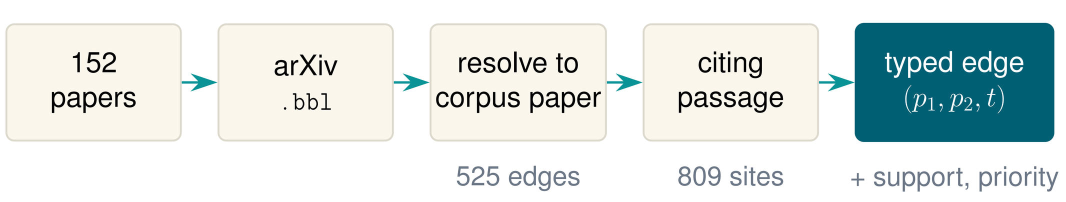


Fig 3. From LaTeX source to a typed edge. The judge sees the citing passage and the cited paper's abstract — nothing else.

4 Building the judge, and the prompt

We reused RefWarden's judge **transport** — the batching, the evidence assembly, the JSON contract — and swapped out the **rubric**. Support and priority came for free; the intent axis was *a prompt change, not new software.* The judge sees the cited paper's abstract as evidence, plus the citing passage with the target marked.

The interesting part is that the constraint inverted.

RefWarden's hard problem is telling the model **what *not* to do** — don't judge the neighbouring citations, don't use prior knowledge, abstain when the abstract doesn't say.

Intent has **no abstain option**: the model must commit to a class on every site. Caution stops being safe — it just becomes a systematic bias toward the majority class.

Which is exactly what happened.

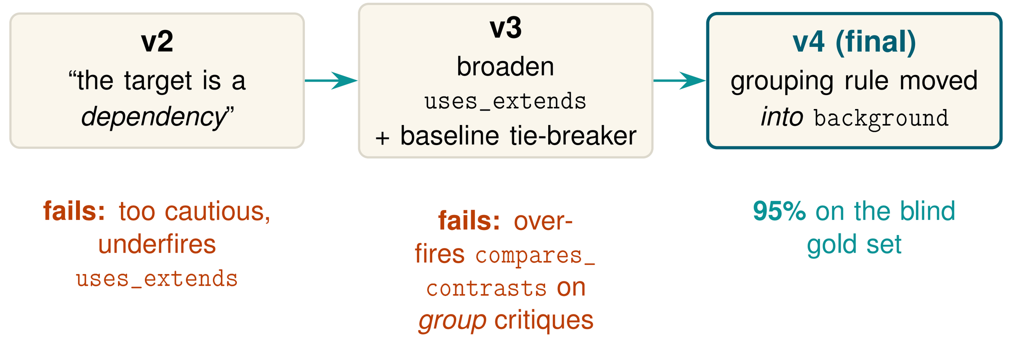


Fig 4. Three rubric versions, each fixing the previous one's failure mode.

- **v2 was too cautious and underfired `uses_extends`.** The rubric was asymmetric: it demanded the target be "a dependency of the citing work" for adoption, while accepting much weaker cues for `background`.
- **v3 broadened it** — the target "provides an ingredient, method, formulation, metric, dataset or architectural block that the citing paper actively uses" — and added a baseline tie-breaker: *rerunning* a system to beat it is `compares_contrasts`, *adopting* its protocol or code is `uses_extends`.
- **v3 then overfired `compares_contrasts` on group critiques.** *"Another limitation is that current systems are limited to small-scale code experiments `[CITED:TARGET]`"* is a critique of a field, not of a paper.
- **v4 fixed it by moving the rule, not adding one.** The "do not over-use `compares_contrasts`" warning came *out* of the negative clause and went *into* the positive definition of `background`: if the target is grouped with other papers to describe a general limitation, it **is** background.

Telling the model what a class *is* beats telling it what a neighbouring class *isn't*.

This was iterative hand-review rather than a controlled ablation — but v4 is the version that was then measured, blind, at 95%.

5 Evaluation

Three validation stages, in increasing order of cost and of trust.

A. Semantic Scholar as a silver standard — a negative result. Free, no LLM. Only 237 of our 525 edges appear in S2's reference graph and **only 10 carry an intent label at all**: SciCite has not been run on 2025–26 papers. On the 9 comparable edges S2 agreed 6/9, and *on all three disagreements S2 was the coarser one.* This is why the next two stages were necessary.

B. A blind second judge. A *stronger* model (Opus 5) relabelled a stratified 100-site sample, given a **clean-room rubric** — it never saw our tie-breakers, so the two annotators' errors decorrelate — and blind to our label.

0.79 Cohen's κ

81% agreement, reweighted

14/14 disagreements on known taxonomy boundaries

Fig 5. Second-judge cross-check — 100 stratified sites, blind.

The result that matters is not the number: **all 14 disagreements land on known taxonomy boundaries, and none are random errors.** A judge that only argues on the genuinely ambiguous cases is a credible judge.

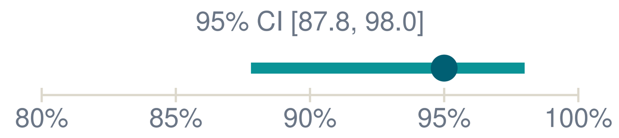
C. A hand-labelled gold set — the real number. 122 sites labelled blind through a purpose-built web app; the judge's verdicts and quality gates sat in the backing sheet but were never sent to the browser. The queue is **blocked by sample type**:

- an 80-site `random_eval` block *in random order*, so any labelled prefix is still a uniform sample and the headline accuracy is unbiased;
- a `disagreement` block of stage-B hard cases, **excluded from accuracy**, used only to referee the taxonomy forks;
- a stratified top-up to $\sim$27 per class for macro-F1.

95.0%

76 / 80 random-block sites

95% CI [87.8, 98.0]



94.6% reweighted to the population class mix
macro-F1 = 0.892

gold

judge			
bg	78.3%	1.3%	2.6%
use	0.3%	6.2%	0.3%
cmp	1.0%	0.0%	10.0%

population-scaled

Fig 6. Headline accuracy with its interval, and the reweighted confusion matrix. Sites on the same edge are not independent, so the true interval is a little wider than binomial.

And the referee result. On the 12 hard cases where our judge and Opus disagreed, gold sided with **ours 6, Opus 5, neither 1**. *The stronger model was not the more accurate one* — and the split was clean: ours won every `background`→X call, Opus won every X→`background` call.

6 Folding citations into the semantic graph

Prior now holds **two graphs** over the same 152 papers, answering different questions. The citation graph is paper→paper and it is a **fact**. The semantic graph is contribution→contribution and it is an **assertion**. Rolling the 989 semantic edges up to paper-pairs and intersecting gives:

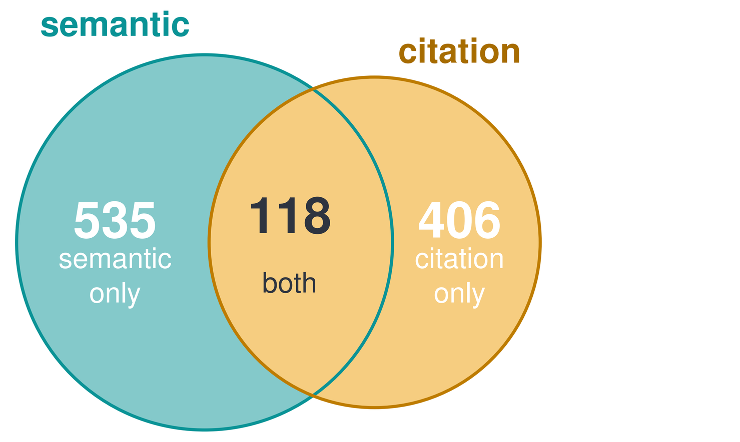


Fig 7. Paper-pair overlap. The two graphs are complementary, not redundant.

Citations alone miss parallel work that never cites; semantics alone invents relations.

The result that justifies the taxonomy: of the 26 shared pairs typed `compares_contrasts`, **only 4** carry any semantic `contradicts` edge. The critique signal is almost entirely invisible to the semantic pass — so the contrast class is not a refinement, it is *new information.*

	supports	builds_on	refines	contradicts
background	57	23	2	8
uses_extends	10	9	1	0
compares_contrasts	19	10	0	4

shared pairs carrying $\geq$1 semantic edge of that relation

Fig 8. Citation intent $\times$ semantic relation, on the 118 shared pairs.

Direction is nearly free. Citation direction agrees with the semantic arrow on **66** pairs and disagrees on **52** — the LLM's `builds_on/refines` arrow is close to a coin flip. On the audited `uses_extends` pairs, citation direction was correct **33/33**.

We then read the divergent pairs by hand, marking each as *complementary*, *semantic-wrong*, *citation-wrong* or *rollup-artifact*. The dominant verdict was **complementary** — the two signals describe different *facets* of a pair rather than disagreeing. The clearest case is MLE-volve→AlphaEvolve, genuinely **both** `builds_on` **and** `contradicts`: it extends the paradigm *and* claims to beat it. A majority-vote rollup would have hidden that tension entirely.

Use citation signal as complementary evidence and as a direction fix — not as a relabel.

Re-typed with citation evidence visible, `uses_extends` pairs became `builds_on` **89%** of the time — without ever overwriting the semantic layer.

[screenshot pending]
`figures/view_intent.png`
Two-layer viewer — both layers on, filtered to the overlap

Fig 9. Both graphs on one layout: solid arrows are citations coloured by intent, dashed lines are the semantic relations.

7 Graph vs. web ideation *(in progress)*

The question: does an LLM exploring the Prior atlas propose better research directions than the same LLM exploring the open web? Same seed topic, same task, same output schema, same generator; only the environment differs. Judged blind and pairwise by a stronger model, order-swapped.

Status: 10 ideas per arm generated; judging not yet built. Three pilot findings are already worth reporting:

1. **Blinding failed first.** 9 of 10 ideas leaked their arm — the graph arm cited internal node IDs, the web arm said "web search". Fixed with an explicit blinding contract and an automated leakage gate before judging.
2. **The arms converged.** 3 of 4 seeds produced the *same core idea* on both sides — unsurprising in hindsight, since the corpus **is** the public literature the web arm retrieves.
3. **The cause: the graph arm wasn't using the graph.** Across four runs it called `get_edges` once and the citation tools zero times — using the atlas as a private search index over the same papers. The fix was not to *force* traversal but to make structure **advertise itself**, surfacing typed-relation breakdowns and citation counts in every tool result. Traversal rose to 3–4 edge calls per run, and 3 of 4 seeds then diverged from the web arm.

	explore calls	seconds	cost / idea
GRAPH	13.6	95	$0.46
WEB	3.9	111	$0.41

Fig 10. Comparable cost per idea — the claim under test is better ideas at similar cost.

8 Takeaways

- **A taxonomy chosen by downstream action, not by convention.** Three intent classes, each kept only because Prior would act differently on it — validated blind at **95% accuracy, macro-F1 0.892**.
- **The rubric was the lever, not the model.** The constraint inverted from "do not classify" to "must classify", and the fix for over-firing was to define a class *positively* rather than forbid its neighbour.
- **The two graphs are complementary** — 118 shared pairs of 1,059 — and the contrast signal is nearly invisible to the semantic layer (4 of 26 pairs). Citation evidence belongs as *evidence and direction*, not as a relabel.
- **Next:** feed intent into the cartographer's labelling prompt and measure whether `contradicts` precision moves off 36%.